School of Computer Science Engineering and Information Systems

Winter Semester 2025-2026

Department of Information Technology

BITE498J – Project – II

Review -1

Smart Placement Email Intelligence System for Students

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ABSTRACT:

It aims to assist students in efficiently managing and analyzing placement-related communications received during campus recruitment. Students receive a large number of emails containing job notifications, eligibility criteria, interview schedules, and attachments, which are often unstructured and difficult to track, leading to missed opportunities and deadlines. This project proposes an intelligent automated system that retrieves placement-related emails using secure email APIs and classifies them into categories such as job notifications, shortlists, interview calls, and offer updates. Essential job-related information, including company name, job role, required skills, eligibility conditions, deadlines, and attached job descriptions, is extracted using Natural Language Processing (NLP) techniques. The system compares the extracted job requirements with the student's stored skill profile and indicate how well the opportunity matches the student's capabilities. The system also detects shortlist emails to verify student eligibility and maintains a detailed placement history, including companies applied, interview rounds completed, and current selection status. A centralized dashboard and reminder system provide timely alerts for deadlines and interview schedules. In addition, a placement-focused community interaction module enables students to share interview experiences, discuss company-specific preparation strategies, and support peer learning.

Keywords: Placement Automation, Email Intelligence, Skill Matching, NLP, Opportunity Tracking, AI-Based Job Analysis.

INTRODUCTION:

Campus placement is a critical phase in a student's academic journey, serving as a bridge between education and professional employment. During this process, students receive a large volume of emails related to job notifications, eligibility requirements, interview schedules, shortlisting updates, and offer letters from various recruiting organizations. These emails are often unstructured, vary in format, and arrive across multiple timelines, making it difficult for students to identify, organize, and prioritize important information effectively. Important deadlines, interview details, or suitable opportunities may be overlooked, negatively impacting placement outcomes. Existing email platforms focus on general communication management and lack intelligent mechanisms to extract, classify, and analyze placement-specific content.

To overcome these challenges, The system automatically retrieves placement-related emails

and applies Natural Language Processing (NLP) techniques to classify them into meaningful categories such as job notifications, interview calls, and shortlisting updates. Additionally, the system maintains placement history, provides timely reminders, and supports placement-focused peer interaction, thereby improving placement readiness and informed decision-making

PROBLEM STATEMENT:

During campus placement processes, students receive numerous placement-related emails containing job notifications, eligibility criteria, interview schedules, and shortlisting updates. These emails are often unstructured and vary in format, making it difficult for students to efficiently track important information. Manual analysis of each email is time-consuming and prone to errors, leading to missed deadlines and overlooked opportunities. Existing email systems lack specialized intelligence to classify placement emails, extract relevant job details, and evaluate student eligibility. Additionally, there is no centralized platform to track placement history or provide personalized insights based on student skills. Therefore, an intelligent system is required to automate the processing, analysis, and organization of placement-related emails to improve placement readiness and informed decision-making for students.

OBJECTIVES:

- To develop an automated system for retrieving and organizing placement-related emails received by students.
- To classify placement emails and extract essential job-related information using Natural Language Processing techniques.
- To compare job requirements with student skill profiles and indicate for better opportunity assessment.
- To verify student eligibility through shortlist detection and maintain a structured placement history.
- To provide a centralized dashboard with reminders and alerts to support efficient placement tracking and informed decision-making
- To offer a centralized dashboard for efficient tracking and management of placement opportunities.

SCOPE OF THE PROJECT:

Predicting whether persons with diabetes are also at risk for heart disease is the primary motive behind this research. Additionally, the goal of this study project is to determine the optimal classification strategy for diagnosing diabetic heart disease in a patient. By employing some classification methods for a comparison study and analysis, this task is supported. Despite the frequent use of these machine learning techniques, accurately predicting diseases remains a significant challenge. This will make it possible for scientists and doctors to develop improvements.

PROPOSED SYSTEM:

The proposed system automatically retrieves and filters placement-related emails using secure email APIs. It classifies emails and extracts essential job details using Natural Language Processing techniques. The system compares job requirements with student skill profiles to generate suitability scores. It verifies student eligibility through shortlist detection and maintains placement history. A centralized dashboard with reminders provides organized tracking and timely alerts for students

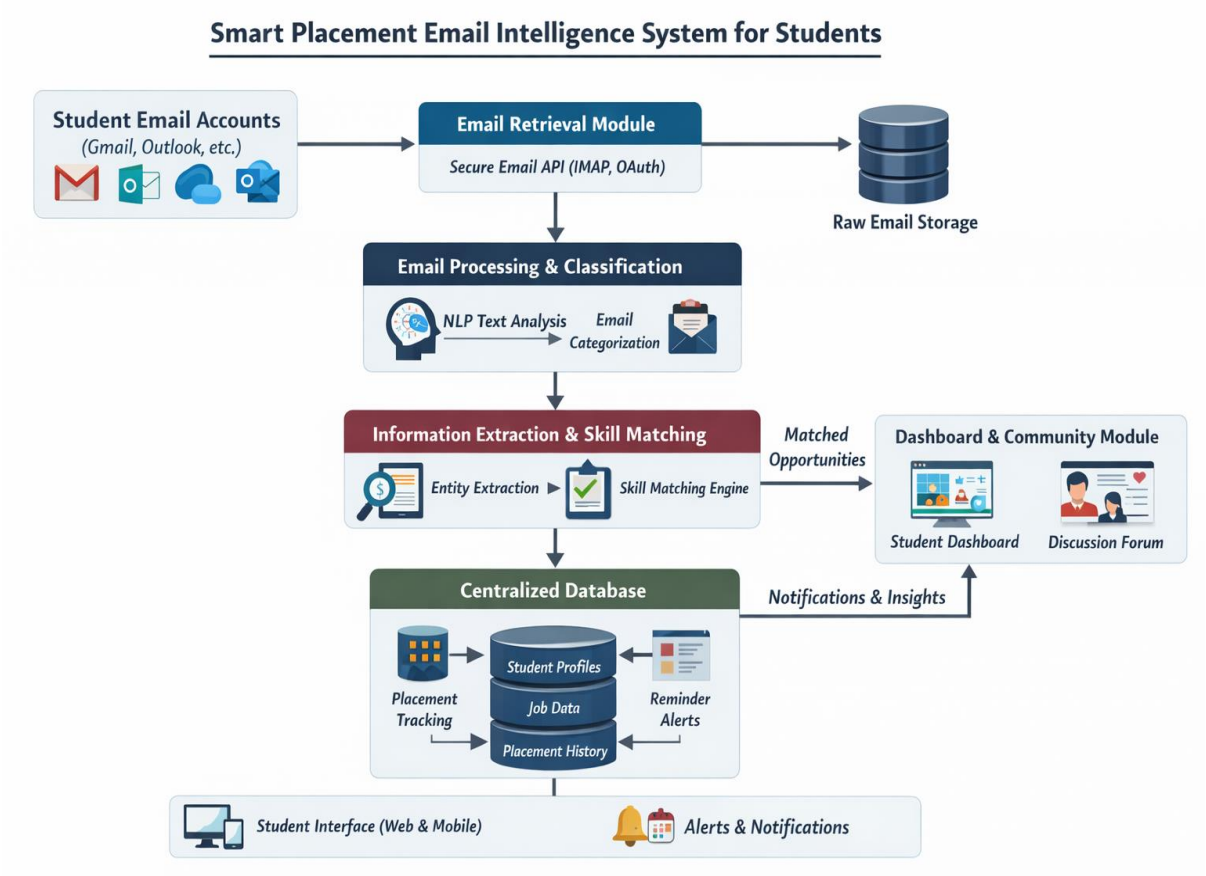
LITERATURE SURVEY:

S. No	Title of the Paper	Merits	Demerits
1	An Intelligent Email Classification System Using Machine Learning	Automatically classifies emails using NLP and ML, reducing manual effort; improves accuracy with more data.	Not tailored for placement emails; does not extract job-specific information like company, role, or eligibility.
2	Automated Information Extraction from Unstructured Emails Using NLP	Efficiently extracts structured information such as dates, entities, and job roles from unstructured emails; reduces manual reading.	Lacks integration with student profiles or skill matching; cannot provide actionable recommendations or alerts.
3	Skill-Based Job Recommendation System Using Text	Matches student skills with job requirements, providing personalized job	Requires manual job data input; does not automate email-based job retrieval or handle live placement

S. No	Title of the Paper	Merits	Demerits
	Mining	recommendations; improves relevance of alerts.	communications.
4	Smart Reminder and Notification System for Event Management	Sends timely alerts and reminders for events and deadlines; ensures tasks are not missed.	Does not analyze email content or extract job-related information; not integrated with eligibility verification.
5	Student Placement Management System Using Web Technologies	Centralizes placement records and tracks student progress; allows administrative oversight.	Relies on manual data entry; lacks AI or NLP for automated email processing or skill matching.
6	NLP-Based Resume and Job Description Matching System	Uses text similarity for accurate skill-job matching; improves relevance of job recommendations.	Cannot process live emails; does not track placement history or generate alerts from incoming communications.
7	Intelligent Dashboard for Academic Decision Support Systems	Provides clear visualization and centralized tracking of student data; supports decision-making.	Focuses on academic data, not placement communication; lacks email processing and skill-based recommendations.
8	Automated Shortlist Detection Using Text Classification	Identifies shortlist emails efficiently; reduces chances of missing eligibility updates.	Limited to shortlisting; does not maintain full placement history or handle other email categories; lacks dashboards and alerts.
9	Email Mining Techniques for Organizational Decision Making	Demonstrates usefulness of email mining to generate structured insights; helps in decision-making.	Not student-focused; lacks personalized job recommendations, dashboards, alerts, or skill matching.
10	AI-Based Career Guidance System for Students	Provides AI-based recommendations to support career decisions; helps students	Does not integrate live placement emails; cannot track student interactions with placement

S. No	Title of the Paper	Merits	Demerits
		choose roles based on skills and trends.	workflows or generate alerts/reminders.

SYSTEM ARCHITECTURE:



1. Presentation Layer (Frontend)

Technologies

- Next.js (React)
- Tailwind CSS
- Web & Mobile responsive UI

Components

- Student Login & Profile Management
- Skill & Resume Upload Interface
- Placement Opportunity Dashboard

- Application Status Tracking
- Notifications & Alerts Panel
- Community / Discussion Forum

Responsibilities

- Displays ML-processed placement opportunities
- Shows matched job recommendations and deadlines
- Sends user actions to backend via REST APIs
- Provides real-time notifications to students

2. Application Layer (Backend Services)

Technologies

- Node.js
- Express.js
- JWT Authentication
- REST APIs

Core Modules

- Authentication & Authorization (JWT)
- User & Student Profile Management
- Email Sync Controller
- Opportunity Management APIs
- Notification Trigger Service
- Community & Messaging APIs

Responsibilities

- Receives requests from frontend
- Manages secure API access
- Coordinates between frontend, ML services, and database
- Sends email content to Intelligence layer for processing
- Delivers ML results back to frontend

3. Intelligence Layer

3.1 Email Classification Model (NLP)

- Classifies incoming emails into:
    - Internship
    - Full-time

    - Coding Test/Interview
    - Rejection / General
- Converts unstructured emails into categorized data

3.2 Information Extraction Engine (NLP – NER)

- Extracts:
    - Company name
    - Role
    - Eligibility criteria
    - Deadline
    - Location
- Uses NLP entity extraction techniques

3.3 Skill Matching & Ranking Model (ML / Hybrid)

- Compares:
    - Extracted job requirements
    - Student skill profiles
- Generates:
    - Match score
    - Ranked opportunity list

3.4 Recommendation Engine

- Personalizes opportunities using:
    - Past applications
    - Skills
    - Deadlines
- Improves relevance for each student

Responsibilities

- Converts raw email data into structured intelligence
- Reduces manual effort for students
- Provides smart recommendations

4. Data Layer

Technology

- MongoDB

Stored Data

- Raw Emails
- Processed Email Metadata
- Student Profiles & Skills
- Placement Opportunities
- Application History
- Notification Logs

Responsibilities

- Stores ML outputs and historical data
- Supports fast retrieval for dashboards
- Enables tracking and analytics

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